



Degree	University/Institute	Year	CGPA/Marks(%)
M.Tech Computer Science & Engineering / NIS - TA	IIT Hyderabad	2026	8.10
BE - Computer Engineering	Madhuben & Bhanubhai Patel Institute of Technology (MBIT)	2024	8.33
XII (GSEB)	P. P. Savani Chaitanya Vidya Sankul	2020	76.00%
X (GSEB)	P. P. Savani Chaitanya Vidya Sankul	2018	93.50%

PROJECTS

Uncertainty Estimation in LLM | *LLM Reliability* | ([Guide: Prof. Srijith P.K.](#)) M.Tech Thesis — Ongoing

- Developing a novel method to estimate uncertainty in large language models (LLMs) **using a single forward pass**, significantly **reducing computational cost** compared to traditional techniques like Bayesian dropout and ensemble methods.
- Extended the **knowledge distillation (KD) framework** for text generation by formulating the teacher as a minimum Bayesian risk aggregator, leveraging multiple model outputs instead of a single teacher view.
- Applied and adapted **skew and reverse-skew KD techniques** to effectively transfer uncertainty-aware knowledge from the teacher ensemble to a single student model.

Motiverse AI 2025 IITH Hackathon | *NLP, RAG, LangChain* ([Github](#)) August - 2025

- Problem 1 – Help-Center Chatbot**: Built a multi-turn RAG chatbot using AWS Bedrock (Claude + Titan embeddings) with **LangChain + FAISS**, added session-aware retrieval (MMR + multi-query), a strict support-agent style to curb hallucinations.
- Problem 2 – Video Indexing and Search**: Built a natural-language video search system: generated per-video descriptors from a feature matrix, indexed them in FAISS for top-K retrieval, and used a **Bedrock (Claude 3.7) + LangChain** agent to interpret queries and filter events by conditions/objects/weather.

MaskCon with Binary-Valued Attributes | *PyTorch, CUB-200-2011 Dataset* ([Github](#)) Course Project-2025

- Integrated binary attribute vectors into the MaskCon framework to enhance representation learning for coarse-grained classes. Engineered and compared three attribute-driven similarity **metrics—cosine, Hamming, and Euclidean**—to guide the masked contrastive loss.
- Optimized training on the **CUB-200-2011 benchmark**, achieving a **Recall@1 boost from 0.5982 to 0.6300 (5.3% relative improvement)**.

Sensor-Based Human Activity Classification | *Custom Transformer, Time-Series CV* ([Github](#)) Course Project — 2025

- Engineered an end-to-end occupancy detection model using raw PIRvision FoG sensor streams—with zero external architectures.
- Architected a bespoke Transformer to **model 55-step temporal sequences** augmented with cyclical time embeddings.
- Devised a **rolling-window cross-validation protocol** to uphold temporal integrity and eliminate data leakage.
- Achieved a 98.2% hold-out accuracy** on real-world FoG data, demonstrating robust generalization.

Building DL Models from the Ground Up | *Core Algorithms, PyTorch* ([Github](#)) Course Mini Project — 2025

- Implemented Perceptron and Gradient Descent, built a **custom MLP from scratch**.
- Trained **CNN, RNN, LSTM, GRU, and ViT models** without external frameworks.
- Created an autograd engine and benchmarked against PyTorch.

INDUSTRY INTERNSHIP EXPERIENCE

Marvell Technology | IITH July-2025

- Profiled Meta's DLRM using PyTorch/Python profilers** to analyze computational & memory bottlenecks, identifying embedding lookups as the dominant runtime overhead.
- Initiated a DAG-based performance model** to estimate **theoretical speedup** under ideal memory conditions, providing actionable insights for future optimization efforts.

SKILLS

Programming Language: C, C++, Python*, Java, Js*

*Elementary Proficiency

Web Tech & Database: HTML, CSS, SQL

Machine Learning: Numpy, Pandas, Sklearn, Tensorflow, PyTorch, Matplotlib, Seaborn, Hugging face

Deep Learning: RNN, LSTM, GRU, Vision Transformer, Autoencoder*

Networking Tools: Wiresharks, tcpdump, mininet*

Core Competencies: Problem-Solving, Effective-Communication, Teamwork, Adaptability, Creativity

SCHOLASTIC ACHIEVEMENTS

GATE CS: Secured an All India Rank of **1823** in GATE Computer Science among **1,23,967** candidates.

GATE DA: Secured an All India Rank of **2751** in GATE Data Science and Artificial Intelligence (DA) among **39210** candidates.

RELEVANT COURSES

Foundations of Machine Learning	Operating Systems	Advanced Computer Networks	Data Structures and Algorithms
Database Management	Discrete Mathematics	Object-Oriented Programming	Probability and Statistics
Computer Organization & Architecture	Deep Learning	Multimedia Content Analysis	Fraud Analytics

EXTRACURRICULAR

Active participant in Clean India Drive by IITH